

Nicholas Mc-Donnell

Software Engineer -- Systems & Performance Engineering

Santiago, Chile

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Experience

Synopsys

Santiago, Chile

Senior R&D Engineer

April 2023 -- August 2026

- Migrated a large C++ production codebase and all its dependencies to a new toolchain and language version.
- Resolved urgent production issues at multiple customer sites, working directly with field engineers.
- Contributed to implementing new geometric-primitive (Bézier curve) support for customer-facing workflows.
- Designed and implemented a debugging/observability layer for the core product, letting issues be instantly captured and reproduced locally instead of requiring custom rebuilds for customers -- significantly cutting debugging and turnaround time for engineering and support.
- Significantly improved task-scheduling performance in the core processing pipeline for complex, large-scale customer workloads.

Personal Projects

Ray Tracer -- Rust

- Implemented a physically based ray tracer following *Ray Tracing in One Weekend*, extending it with an experimental BVH acceleration structure and multithreaded rendering.
- Restructured the rendering pipeline to enable tail-call optimization and improve the architecture for future performance work.
- Currently exploring Rust based GPU-accelerated rendering.

Segment Tree Library -- Rust

- Designed and implemented a generic, high-performance segment tree library supporting range queries, range updates, and lazy propagation.
- Designed the API for flexibility across different segment-tree use cases while maintaining efficient implementations.
- Currently implementing wide-tree layouts to improve cache locality and performance.

Teaching and Research

Millenium Institute Foundational Research on Data

Santiago, Chile

Software Engineer

September 2020 -- December 2022

- Worked in projects about machine learning and interpretability.

Computer Science Department (PUC)

Santiago, Chile

Teacher Assistant

March 2021 -- December 2022

- Designed homework assignments and supplementary problems emphasizing algorithmic problem solving and non-obvious applications of techniques introduced in class.

Courses:

- Cryptography
- Algorithms & Data Structures
- Algorithm Design & Analysis

Mathematics Faculty (PUC)

Santiago, Chile

Teacher Assistant

March 2018 -- December 2020

- Led problem-solving sessions for undergraduate mathematics courses, working through exercises and guiding students through rigorous solution strategies in real time.

Courses:

- Real Analysis
- Abstract Algebra
- Several Calculus Courses

Education

Pontificia Universidad Católica de Chile (PUC)

Santiago, Chile

Mathematics B.Sc.

March 2017 -- December 2020

Coursework: Algorithms & Data Structures, Algorithm Design & Analysis, Probability & Statistics, Linear Algebra, Algorithms, Abstract Algebra

Activities: Model Theory Seminar Organizer, Mathematics Student Council, Mathematics Statutes Commission

Skills

Programming Languages: C++, Rust, Python, Java, SQL, Bash/Zsh

Tools: Linux, Git, Jujutsu, Vim, VSCode, Zed

Languages: Spanish (Native), English (professional/fluent)

Awards

International Collegiate Programming Contest Latin American Regional 2nd place in Chile (54th in Latin America)	2019
IEEEEXTREME 15.0 26th place	2021
IEEEEXTREME 16.0 23rd place	2022